

****Document Control****

- * ****Project Name:** PadelSync**
- * ****Document Type:** UX/UI Design Strategy & Screen Requirements Brief**
- * ****Author:** UX/UI Team Lead**
- * ****Target Audience:** Product Owner, Lead Architect, UX/UI Design Team, Flutter Engineering**
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1. Design Vision & Core Principles

Our goal is to create an interface that feels like a premium, trusted sports companion. Because Padel is a fast-paced, social sport often played in varied lighting conditions (bright outdoor sun or dim indoor/basement courts), our UI must prioritize ****legibility, high contrast, and one-handed usability****.

****Core UX Principles:****

1. ****Offline Transparency:**** Users must **never** guess if their data is saved. The UI must clearly communicate local vs. cloud states without causing anxiety.
2. ****Frictionless but Exclusive:**** The registration process should feel easy, but the "Pending Approval" state should reinforce the exclusivity and trust of the vetted community.
3. ****Optimistic Interactions:**** The app should feel instantaneous. We will design for "Optimistic UI" where the interface updates immediately upon user action, syncing with Firebase in the background.
4. ****Material Design 3 (Android First):**** Since the MVP is Android-native (via Flutter), we will strictly adhere to Material 3 guidelines for navigation, haptics, and dynamic color theming.

2. Core User Journeys (Flows)

Flow A: The Unregistered Player (The "Vetting" Journey)

1. **Splash & Value Prop:** Quick onboarding carousel explaining the "Vetted Community" concept.
2. **Request Access Form:** Input Name, Email, Phone, and "Home Club" (Dropdown fetched from cached Admin locations).
3. **The Waiting Room:** A visually engaging "Pending Approval" screen. *UX Note: Include a "Share Invite Link" button so they can invite their regular padel partners, which helps Admins verify them faster.*
4. **Activation:** Push notification -> Deep link directly to Profile Setup -> Home Dashboard.

Flow B: The Active Player (The "Match" Journey)

1. **Dashboard:** View upcoming matches, quick stats, and recent history.
2. **Create Match:** Select Date/Time -> Select Location -> Select 3 Partners/Opponents from the vetted roster.
3. **Play & Log:** Post-match, open the match card. Use a custom **Score Stepper** component to log sets (e.g., 6-4, 3-6, 6-2).
4. **Sync & Share:** View the "Syncing..." badge transition to "Saved". One-tap share to Instagram/WhatsApp with a generated graphic of the match result.

Flow C: The Admin (The "Management" Journey)

1. **Admin Dashboard:** Hidden tab that only appears for users with the `role: admin` custom claim.
2. **Triage Queue:** Swipeable cards for "Pending Users" (Swipe Right to Approve, Left to Reject).
3. **Location Management:** Map view to drop pins on new local courts.
4. **Dispute Resolution:** View flagged matches with conflicting scores and override the final result.

3. Screen-by-Screen Requirements (The "What")

| Screen Name | Key Elements & Data Requirements | Firebase/Architectural Tie-in |

| :--- | :--- | :--- |

| **Home Dashboard** | - Greeting & Avatar
- "Next Match" Card
- Quick Stats (Win %)
- Recent Match List | Uses Firestore real-time listeners (` snapshots() `) to auto-update if a partner creates a match. |

| **Match Creation** | - Date/Time Picker
- Location Selector (Autocomplete)
- Player Roster (Searchable list of *Active* users only) | Must gracefully handle offline state. If offline, disable live-search and show the locally cached roster. |

| **Score Entry** | - Custom Set Score Stepper
- "Finalize Match" CTA
- Sync Status Badge | **Optimistic UI:** Clicking "Finalize" instantly locks the UI and shows "Saving...". |

| **Match Details** | - Court Location Map
- Player Names & Avatars
- Scoreboard
- "Dispute Score" button | If ` status == DISPUTED ` , overlay a "Under Admin Review" banner and disable editing. |

| **Admin: Triage** | - List of Pending Users
- Registration Date
- Mutual Connections (e.g., "Plays with [Active User]") | Real-time stream. Swiping triggers a Cloud Function to update Auth Custom Claims. |

| **Admin: Locations** | - List of Clubs
- "Add New" FAB (Floating Action Button) | "Add New" triggers Google Places API search before saving to Firestore ` locations ` collection. |

4. UX Patterns for Firebase & Offline-First (Crucial)

To align with the Lead Architect's Firebase Blaze strategy, the design team must implement the following specific UX patterns:

A. The "Sync Status" Indicator System

We need a global or contextual component to show data state.

- *  **Cloud Synced:** Subtle checkmark icon. Fades out after 3 seconds to reduce clutter.
- *  **Saving Locally / Syncing:** Animated spinner or pulsing dot. Text: "Saving to device..." or "Syncing to cloud..."
- *  **Sync Failed:** Red warning icon. Text: "Offline. Saved to device. Tap to retry sync..."

B. Optimistic UI for Score Logging

When a user inputs a score and hits "Save":

1. **Instant Feedback:** The UI immediately updates the Win/Loss stats on the dashboard.
2. **Background Process:** The Sync Status badge changes to  *Syncing*.
3. **Completion:** Once Firebase acknowledges the write, the badge changes to  *Synced*.
4. **Conflict Handling:** If the backend rejects the write (e.g., the other player already submitted a different score), the UI rolls back, shows a  *Dispute Flagged* toast notification, and locks the match card.

C. Empty States & Skeleton Loading

- * **Skeleton Screens:** Use shimmering skeleton loaders instead of spinning wheels when fetching the match history.
- * **Offline Empty States:** If a user has no local cache and opens the app in Airplane Mode, show a friendly illustration: "You're offline and we have no cached matches. Connect to the internet to load your history..."

5. Design System & Component Needs

To ensure consistency and speed up Flutter development, we will build a robust Figma Design System including:

* **Color Palette:**

* **Primary:** Padel Court Blue (` #0055FF `) and Artificial Grass Green (` #00C853 `).

* **Backgrounds:** High-contrast Dark Mode (OLED black ` #121212 `) to save battery and look premium; Light Mode must use off-white (` #F8F9FA `) to reduce glare in outdoor sun.

* **Typography:** Geometric sans-serif (e.g., *Inter* or *Montserrat*) for high legibility of numbers/scores.

* **Custom Components to Design:**

* `ScoreStepper` : Large touch targets for incrementing/decrementing set scores (e.g., 0 to 7).

* `PlayerChip` : Compact avatar + name + "Online/Offline" indicator for match creation.

* `SwipeActionCard` : For Admins to approve/reject users.

* `SyncBadge` : The 3-state indicator mentioned above.

6. Visual North Star (UI Concept)

To align the team visually, I have generated a high-fidelity concept of the **Score Entry Screen** showcasing the offline sync indicator and dark mode aesthetic.

(Generating Visual Concept...)

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- * ****Date:** May 23, 2026**

1. Design Vision & Core Principles

As the UX/UI Team Lead, my goal is to translate the Product Owner's vision and the Lead Architect's technical framework into a seamless, intuitive, and reliable user experience.

Our design will be guided by three core principles:

1. ****Offline Transparency:**** Because the app relies on Firebase's offline-first architecture, users must never feel anxious about losing data. The UI must clearly and calmly communicate what is saved locally versus what is synced to the cloud.
2. ****Outdoor Readability & Ergonomics:**** Padel is played outdoors in bright sunlight or in brightly lit indoor clubs. Players will often use the app with sweaty hands or while in a rush between sets. We will prioritize high-contrast typography, large touch targets, and a "Light Mode" default optimized for glare.
3. ****Community Trust:**** The "vetted" nature of the app should feel exclusive and secure, not bureaucratic. The onboarding flow must feel premium and welcoming.

(Note: I have generated a high-fidelity UI concept above to serve as our "Visual North Star" for the design team, showcasing the modern, high-contrast aesthetic and the offline sync indicators we will be implementing).

2. Core User Journeys (Flows)

Journey A: The Gatekept Onboarding (Unregistered → Active Player)

1. **Splash & Auth Gate:** User opens the app. If no token, show "Welcome to PadelSync" with a "Request Access" CTA.
2. **Registration Request:** Simple form (Name, Email, Phone, Home Club).
3. **The Waiting Room:** A beautifully designed empty state explaining that an Admin is reviewing their request. Includes an estimated wait time and a "Pull to refresh" mechanic.
4. **Approval & Onboarding:** Push notification received. User opens app, sees a "Welcome!" animation, and is taken through a 3-step tutorial on how to log their first match.

Journey B: The Match Lifecycle (Active Player)

1. **Dashboard:** Shows upcoming scheduled matches, recent results, and a quick "Log Match" FAB (Floating Action Button).
2. **Match Creation:** Select Date/Time → Select Location (from cached list) → Select 3 other players (Team A vs Team B).
3. **Score Entry:** A specialized "Score Stepper" UI for entering sets (e.g., 6-4, 3-6, 6-2).
4. **Post-Match:** Summary screen showing the new Win/Loss impact on their profile stats.

Journey C: The Community Manager (Admin)

1. **Admin Dashboard:** A dedicated tab (only visible to Admins) with three sections: Pending Users, Location Management, and Dispute Resolution.
2. **User Approval:** Tinder-style swipe or simple Accept/Reject buttons on pending user cards.
3. **Dispute Resolution:** View the disputed match, see the conflicting scores submitted by both teams, and manually override the final result.

3. Screen-by-Screen Requirements

| Screen Name | Key Elements & Interactions | Technical/UX Constraints |

| :--- | :--- | :--- |

| **Registration Request** | Form fields, Home Club dropdown (searchable). | Must validate email format locally before submission. |

| **Pending Approval** | Animated illustration, "Check Status" button. | Must handle Firestore real-time listeners to auto-transition to the Dashboard the second the Admin approves them. |

| **Home Dashboard** | "Next Match" card, "Recent Results" list, Global Stats summary. | Must load instantly from local cache. Skeleton loaders for remote data. |

| **Match Creation** | Multi-step wizard or single scrollable page. Player search with autocomplete. | Must work 100% offline. Player search must query the local cached `user\_profiles` collection. |

| **Score Entry** | Custom Padel Scoreboard UI. Set toggles (Set 1, Set 2, Set 3). Tie-break toggles. | Needs haptic feedback on score changes. Must prevent illogical scores (e.g., 6-1, 6-0, 6-0). |

| **Admin Dashboard** | Tabs for Users, Locations, Disputes. | Hidden via RBAC. Must use swipe-to-dismiss for approving/rejecting users. |

| **Player Profile** | Avatar, Win/Loss pie chart, Match History timeline. | Stats are read from the pre-calculated `user\_stats` document. |

4. UX Patterns for Firebase & "Offline-First" (Crucial)

This is where UX meets Architecture. Because we are using Firebase Firestore's offline persistence, the UI must handle state changes gracefully without blocking the user.

A. The "Optimistic UI" Pattern

When a user logs a match score while offline (or on a slow connection), we will not show a blocking "Loading..." spinner.

- * **Action:** User taps "Save Score".

- * **Immediate UI:** The screen transitions to the Match Summary, and the score is displayed immediately. A subtle **"Saving locally..."** badge appears at the top of the card.

- * **Background Sync:** Firestore writes to the local cache instantly, then syncs to the cloud.

- * **Resolution:** Once the cloud acknowledges the write, the badge smoothly transitions to a green **"Synced"** checkmark. If the sync fails after multiple retries, it turns red with a **"Tap to Retry"** action.

B. Real-Time Listeners (No "Pull to Refresh" needed)

For the Admin "Pending Users" queue and the Player "Match History", we will use Firestore's real-time `snapshot()`.

- * **UX Impact:** When a new user registers, they will literally animate into the Admin's list in real-time without the Admin needing to pull down to refresh. This makes the app feel incredibly alive and responsive.

C. Handling Disputes (The "Locked" State)

If a match score is disputed, the UI must clearly prevent further edits.

- * **Visual Cue:** The match card will feature a yellow "Under Review" banner. The "Edit Score" button will be disabled and grayed out.

- * **Admin View:** The Admin will see a side-by-side comparison of what Player A submitted vs. what Player B submitted, with a modal to input the "Official Final Score".

5. Design System & Component Needs

To ensure consistency and speed up Flutter development, the design team will build a comprehensive Figma Design System including:

* **Color Palette:**

* **Primary:** Padel Court Blue (Vibrant, high contrast).

* **Secondary:** Neon Green (for success states, active indicators).

* **Backgrounds:** Off-white/Light Gray (to reduce eye strain in bright sunlight).

* **Typography:** A geometric, highly legible sans-serif (e.g., *Inter* or *Montserrat*). Large numeric fonts for scores.

* **Custom Components to Design:**

1. **Sync Status Badge:** (States: Saving, Synced, Offline, Error).

2. **Padel Score Stepper:** A custom widget for incrementing/decrementing sets and games, including a toggle for "Tie-break".

3. **Player Chip:** A compact UI element showing a player's avatar, name, and a small "Verified" badge, used heavily in the Match Creation flow.

4. **Swipe-to-Approve Card:** For the Admin user queue.

6. Accessibility & Edge Cases

* **Sunlight Mode:** Ensure all text meets WCAG AAA contrast ratios against the background.

* **Fat Finger Syndrome:** All interactive elements (buttons, list items) must have a minimum touch target of 48x48dp.

* **Empty States:** Design engaging illustrations for when a user has no matches yet, or when an Admin has no pending approvals (e.g., "All caught up! Time to hit the courts.").

7. Next Steps & Deliverables Timeline

To align with the Engineering team's Sprint 0, the UX/UI team will execute the following plan:

1. **Week 1: Low-Fidelity Wireframes & User Flows**

- * Deliverable: Clickable grayscale prototype in Figma to validate the flows with the PO.

2. **Week 2: High-Fidelity UI & Design System**

- * Deliverable: Finalized screens, color palette, typography, and custom component library.

3. **Week 3: Interactive Prototype & Dev Handoff**

- * Deliverable: Fully interactive Figma prototype demonstrating the "Optimistic UI" sync states and Admin swipe gestures.

- * **Handoff:** We will use Figma's Dev Mode to provide the Flutter engineers with exact spacing, colors, and asset exports, alongside a documented guide on how the Sync Status Badges should behave.

PO & Architect: Please review this UX strategy. If the approach to the "Offline-First" UX patterns and the Admin flows aligns with your expectations, my team will begin wireframing the Registration and Match Creation flows immediately.